Takuo Yamamoto

Brooklyn, New York — (929)406-9935 — takuoymt51@gmail.com

Education

Brooklyn Technical High School **Expected June 2027**

Aerospace Engineering Major; New York State Advanced Regents Diploma

Japanese Sunday Waldorf School **2016 – 2024**

Bilingual Education & Cultural Studies

- Dedicated 8 years to weekly bilingual education, studying Japanese literature, history, and cultural heritage within a collaborative community environment.

Research & Independent Projects

MIT GEAR Lab **Summer 2026**

Research Assistant, Heavy-Duty Vehicle Electrification

- Modeled refuse fleet energy consumption and payload tradeoffs across electric and hydrogen powertrains using drag, rolling resistance, and duty cycle metrics.
- Developed a web calculator comparing electric, hydrogen, and diesel powertrains based on vehicle operating conditions.

Author, Published Literature Review **Sept. 2025 – June 2026**

Engineering & Economics Feasibility of Sustainable Aviation Fuel (SAF)

- Conducted synthesis on SAF engineering and economic feasibility under mentorship from Yonglin He, Stanford graduate and Lockheed Martin Engineer.
- Evaluated energy density, lifecycle emissions, production pathways, scalability, and commercialization; published research review. doi.org/10.5281/zenodo.21501806

Experience

Brooklyn Technical High School, Tech News Network (TNN) **Nov. 2025 – Present**

Co-founder, Executive Producer & Outreach Coordinator

- I lead operations for a student-run news network with 20+ members reaching 250,000+ viewers; manage production workflows, scheduling, communications, and led outreach to establish partnerships with Columbia GS of Journalism.

Japan Karate Association, Shotokan Karate-Do International **2017 – Present**

Instructor, National Athlete

- 9-year practitioner, 1st Dan black belt. Teach beginner classes of approximately 15 students and mentor younger students.US Nationals Competitor.

Sixth Sense: An AI Assistive Navigation Research Project **Feb. 2026 – Present**

Project Developer

- Co-created a wearable system for the visually impaired using Jetson Orin Nano, stereo vision, YOLO & OpenCV. BTHS Hackathon winner, Griptape Challenge participant.

Brooklyn Boatworks, NYC Youth STEM Nonprofit **Nov. 2024 – 2026**

Junior Leadership, Camp Counselor-in-Training (CIT)

- Taught students woodworking; led summer camp activities/sailing; Opti-launch volunteer; captained 1st-place team at annual PioneerWorks Boat Regatta.

Brooklyn Bridge Marina **Aug. 2026 – Present**

Marina Operations Apprentice, Dockhand

- Supporting guest services, vessel operations, sailing programs, events, and dock maintenance. Coordinated VIP visits, maintenance of Melges 24s & J/80s, and upkeep of the marina.

Clark Index **Sept. 2025 – Present**

Co-Founder, Website Developer

- Built a website platform with 75+ users using logistic regression to forecast games; explained prediction factors to make football more accessible.

Summer Study Abroad, CIEE Toulouse **Summer 2025**

Study Abroad Student

- Studied aerospace fundamentals, aerodynamics, and rocketry; engaged with professors and professionals from Airbus and ENAC.

Exceed Microwave **Summer 2024**

Internship

- Supported basic manufacturing and testing of precision electronic filters for defense and aerospace customers, gaining experience in calibration and quality assurance.

Awards & Certifications

1st Place, DIY Category | *Pioneer Works NYC, Red Hook Boat Regatta* May 2026
1st Place, International Individual Kata | *46th Annual JKA SKDI* Spring 2026
Aerospace Work Experience Certification | *In partnership with Airbus (Springpod)* July 2025
Shodan (1st Dan) Black Belt | *Shotokan Karate* Jan. 2023

Technical Skills & Languages

Skills: Autodesk Inventor, Technical Writing, Team Management, Electronic Testing & Calibration
Languages: English (Native), Japanese (Native speaking/writing, Intermediate reading), Spanish (Intermediate)
Volunteer Activity: NYC TCS Marathon Volunteer (3 years), Brooklyn Japanese American Family Association Volunteer (3 years)